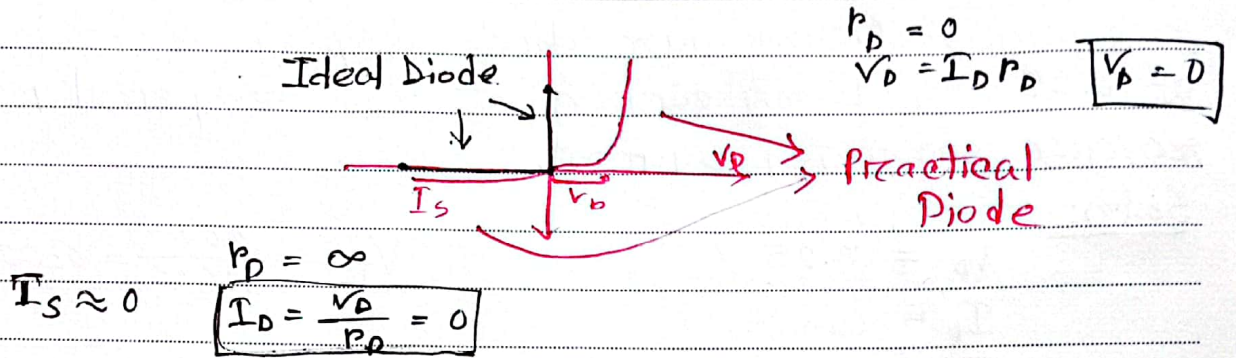


Ideal v/s Practical Diode



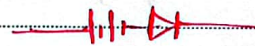
Equivalent Circuit for Diode

* Piecewise-Linear Eq. C:

कार्योति, एकांकिक, त्रुति अरु धातु (कम) रस।



* Simplified: अर्ध कार्योति + एकांकिक



We use it

* Ideal:



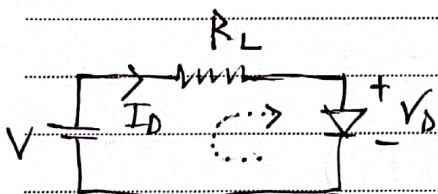
Reverse Bias



Forward Bias

Load Line Analysis

Non linear or Ohm's Law भन्ने हुन न



Applying KVL:

$$V - I_D R_L - V_D = 0$$

$$\Rightarrow V = I_D R_L + V_D \quad \text{--- (i)}$$

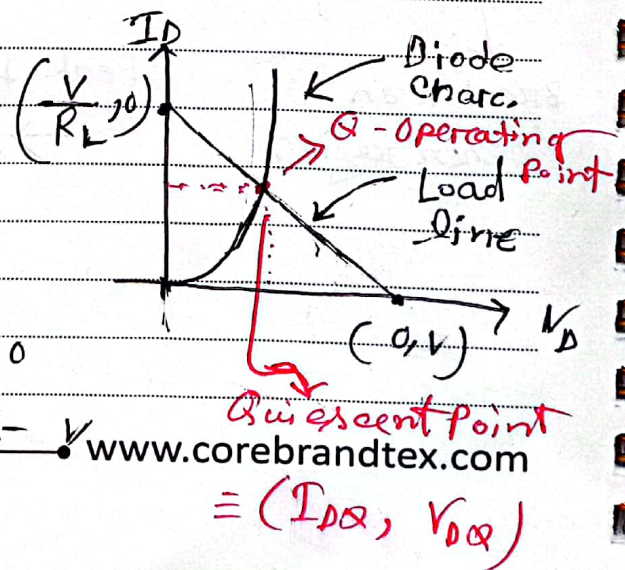
When,

$$V_D = 0 \Rightarrow V = I_D R_L + 0$$

$$I_D = V / R_L$$

$$I_D = 0$$

$$\Rightarrow V_D = V$$



$$V = I_D R_L + V_D \Rightarrow \frac{V}{R_L} = I_D = \frac{V_D}{R_L} \quad \text{or, } I_D = \frac{-V_D}{R_L} + \frac{V}{R_L}$$

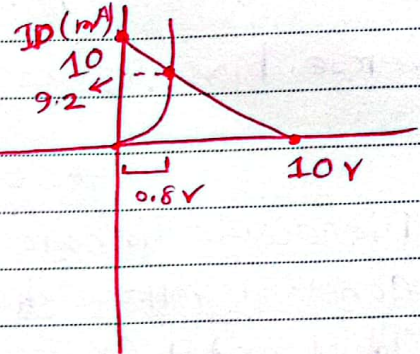
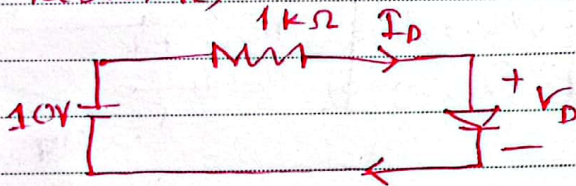
$$y = mx + c$$

Slope: $c = \frac{V}{R_L}$

Slope (m) = $-\frac{1}{R_L}$

Problem

Determine Q-pt and V_R (Voltage drop across resistor)



$$V_D = 0.8V$$

$$V = 10V$$

$$R_L = 1k\Omega$$

Using KVL,

$$10 - 1 \times 10^3 I_D - V_D = 0$$

$$\Rightarrow 10 = 1 \times 10^3 I_D + V_D$$

When, $V_D = 0 \rightarrow I_D = 10 \text{ mA}$

$I_D = 0 \rightarrow V_D = 10$

$$\therefore Q_{pt} = (I_{DQ}, V_{DQ}) = (9.2, 0.8)$$

Ans.

$$\therefore V_R = I_D R_L = 9.2 \times 10^{-3} \times 1 \times 10^3 = 9.2V$$

(Ans.)

Resistance of a diode

Forward Bias G,

$$\text{DC resistance, } R_f = \frac{\text{DC voltage}}{\text{DC current}}$$

$$\text{Dynamic / AC resistance, } r_f = \frac{d(\text{Voltage})}{d(\text{current})}$$

$$\text{Average resistance} = \frac{\Delta V}{\Delta I}$$

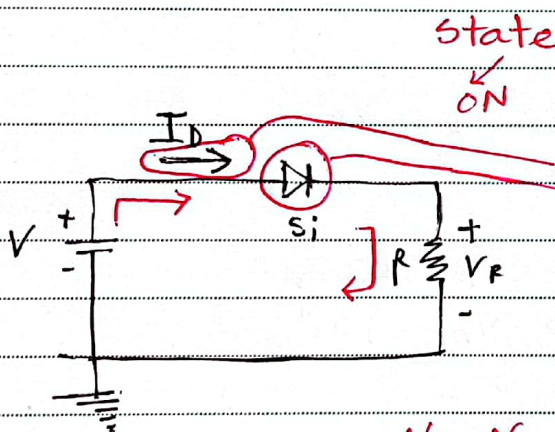
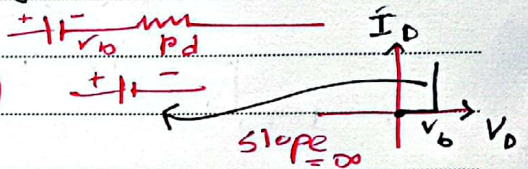
Reverse bias G, resistance MΩ লক্ষকোটি।

Series Diode Configurations

(i) Piecewise linear model. (V_b, r_d)

(ii) Constant voltage drop model. ($V_b, r_d = 0$)

(iii) Ideal model. ($V_b = 0, r_d = 0$)



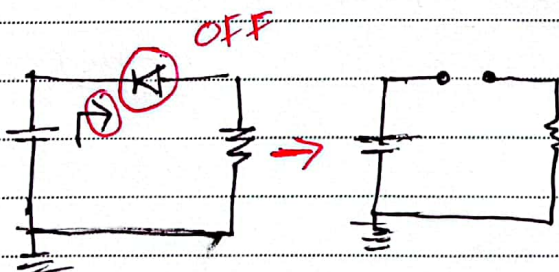
Direction of current যদি Arrow of diode এর গিছে দিলে হয় তবে Diode = ON

$$V - V_K - V_R = 0$$

[KVL]

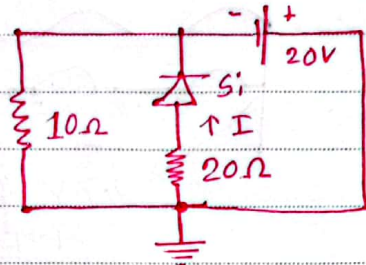
$$\Rightarrow V_R = V - V_K$$

$$I_D = I_R = \frac{V_R}{R}$$

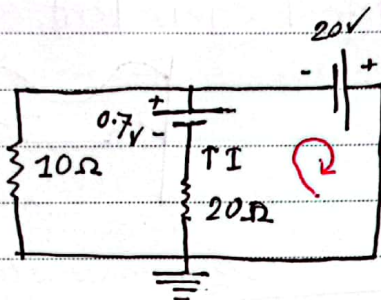


$$V_R = 0$$

Example: Determine the current I in the configuration shown using constant voltage drop model.



Diode is on.



Applying KVL,

$$20V - 20I - 0.7V = 0$$

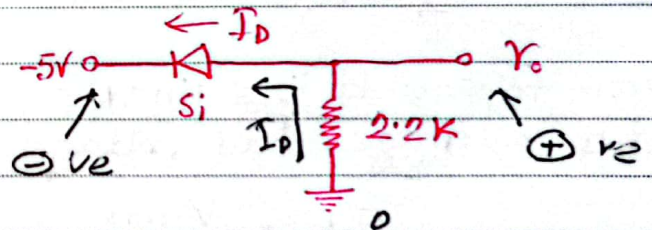
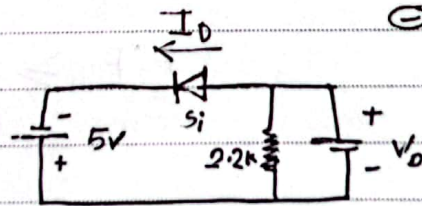
$$\Rightarrow 20I = 20V - 0.7V$$

$$\Rightarrow I = 965 \text{ mA}$$

(Ans.)

Series Diode Configuration (Examples)

① Determine V_o and I_D .



$$\star V_o = -5V + 0.7V = -4.3V$$

$$\star (KVL) \Rightarrow 0V - 22I_D = V_o$$

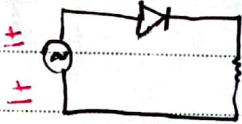
$$\Rightarrow I_D = \frac{4.3V}{22k\Omega} = 1.955 \text{ mA}$$

(Ans.)

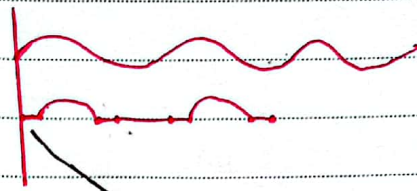
Rectifier

- Half wave
- Full wave
 - center tap
 - Bridge type

ON OFF Half wave rectifier



Piecewise equivalent circuit:



0.7V এর কারণে (+) হলে
আমরা 0.7V লিখি

Ideal equivalent circuit

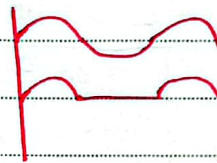


fig: characteristics of half w. rr.

$$* V_{av} = \frac{V_m}{\pi} = 0.318 V_m$$

$$* I_{av} = \frac{V_{av}}{R} = \frac{V_m / \pi}{R}$$

$$= \frac{V_m}{R\pi} = \frac{I_m}{\pi}$$

$$* I_{rms} = \frac{I_m}{2}$$

$$* V_{rms} = I_{rms} \cdot R$$

$$= \frac{V_m}{2R} \times R$$

$$= \frac{V_m}{2}$$

Form factor: It is defined as the ratio of rms load voltage and average load voltage.

$$F.F = \frac{V_{rms}}{V_{av}}$$

$$= \frac{V_m / 2}{V_m / \pi}$$

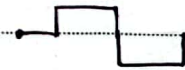
$$= \frac{\pi}{2}$$

$$= 1.57$$

$$V_{rms} = 1.57 \cdot V_{av}$$

$$F.F \geq 1$$

$$r_{ms} \geq av$$



এরকম আকৃতি

$$V_{rms} = V_m$$

$$V_{av} = V_m$$

$$\therefore FF = 1$$

Ripple Factor: The output current contains both ac and dc components. The ripple factor measures the percentage of ac component in the rectified output.

The ideal value of ripple factor should be zero. (0%)

$$\gamma = \frac{\text{rms value of ac component of output}}{\text{avg value of output}}$$

$$I_{ac \text{ rms}} = [I_{rms}^2 - I_{dc}^2]^{1/2}$$

$$\therefore \gamma = \left[\frac{I_{rms}^2}{I_{dc}^2} - 1 \right]^{1/2} = \sqrt{(F.F)^2 - 1}$$

$$= \sqrt{1.57^2 - 1} = 121\%$$

↑
এটা অসম্ভব মান

Efficiency: It is defined as ratio of dc power available at the load to the input ac power.

$$\eta\% = \frac{P_{load}}{P_{in}} \times 100\%$$

$$P = VI$$

$$P = I^2 R$$

$$= \frac{I_{dc}^2 R}{I_{rms}^2 R} \times 100\%$$

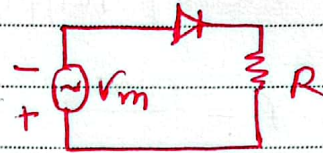
$$I_{dc} = I_m / \pi$$

$$I_{rms} = I_m / 2$$

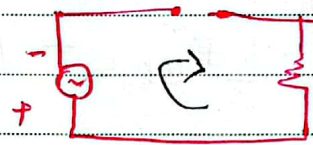
$$= \frac{\pi^2}{4} \times 100\% = 40.56\% \leftarrow \text{এটা অসম্ভব মান}$$

Peak Inverse Voltage : Maximum reverse bias voltage that can be applied across the diode before entering to the Zener or breakdown region.

$$PIV = V_m$$



$$P \geq V_m \text{ (প্রমাণিত (অনুস))}$$



$$-V_m + V = 0$$

$$V = V_m$$

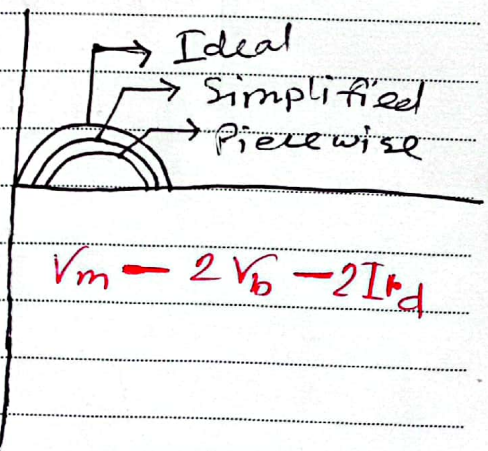
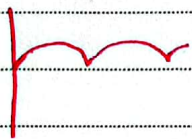
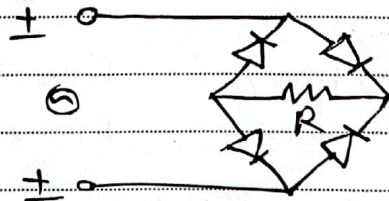
Full wave bridge rectifier

Output up to 100%

Half w.r. i) $\oplus$ ve half cycle $\rightarrow V_o = V_i = V_m \sin \omega t$
 $\ominus$ ve $\dots \rightarrow V_o = 0$

Disadvantages of H.W.R. ii) $\gamma\% = 42.1\%$ — Ripple factor
 iii) $\eta\% = 40.56\%$ — Efficiency

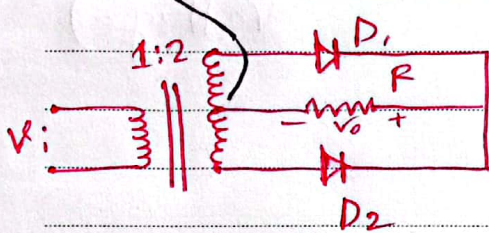
F.W.R :



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Center tapped

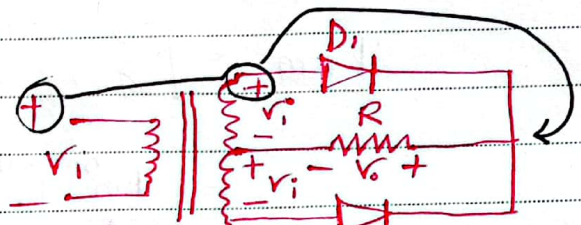
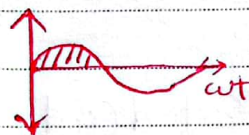
Full wave center-tapped rectifier



क्या मजिदें वृत्त वृत्त (आवृत्त) आवृत्त चारे

$$\frac{N_P}{N_S} = \frac{V_P}{V_S} \Rightarrow \frac{1}{2} = \frac{V_P}{V_S} \Rightarrow V_S = 2V_P$$

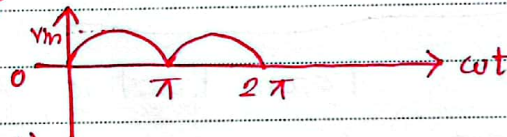
Positive half cycle:



Similarly, $\ominus$ ve वृत्त आवृत्त D_2 फिरे मारु ।

Full Wave Rectifier (Avg. Load Current & Avg. Load V.)

Average (dc) load voltage:



$$\begin{aligned} V_{av} = V_{dc} &= \frac{1}{\pi} \int_0^{\pi} V_m \sin \omega t \, d(\omega t) \\ &= \frac{V_m}{\pi} [-\cos \omega t]_0^{\pi} \\ &= \frac{V_m}{\pi} [-\cos \pi - (-\cos 0)] \end{aligned}$$

$$V_{av} = \frac{2V_m}{\pi}$$

Average (dc) load current:

$$I_{av} = \frac{V_{av}}{R} = \frac{2V_m}{\pi R}$$

$$I_{av} = \frac{2I_m}{\pi}$$

$$I_{av} = \frac{1}{\pi} \int_0^{\pi} I_m \sin \omega t \, d(\omega t)$$

www.corebrandtex.com

Full Wave Rectifier (RMS Load Current and RMS Load Voltage)

RMS Load Current:

I = load current

$$I = I_m \sin \omega t \quad 0 \leq \omega t \leq \pi$$

$$I_{rms} = \left[\frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \omega t \, d(\omega t) \right]^{1/2}$$

$$= \left[\frac{I_m^2}{\pi} \int_0^{\pi} \sin^2 \omega t \, d(\omega t) \right]^{1/2}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

RMS Load Voltage:

$$V_{rms} = I_{rms} R = \frac{I_m}{\sqrt{2}} \times R$$

Form Factor & Ripple Factor

Form Factor:

$$f.f = \frac{V_{rms}}{V_{av}} \quad \text{or,} \quad \frac{I_{rms}}{I_{av}}$$

$$= \frac{V_m / \sqrt{2}}{2 V_m / \pi}$$

$$= \frac{\pi}{2\sqrt{2}}$$

$$= 1.11$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$V_{av} = \frac{2 V_m}{\pi}$$

Ripple Factor: $\gamma = \sqrt{1.11^2 - 1} \times 100\%$

$$= 48.1\%$$

ac component

Efficiency and PIV

Efficiency:

$$\eta\% = \frac{\text{o/p dc power}}{\text{i/p ac power}} \times 100\%$$

$$= \frac{I_{dc}^2 R}{I_{rms}^2 R} \times 100\%$$

$$= \frac{4 I_m^2 / \pi^2}{I_m^2 / 2} = \frac{8}{\pi} \times 100\%$$

$$P = I^2 R$$

$$I_{dc} = \frac{2 I_m}{\pi}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

$$\eta\% = 81.13\% = \text{Double of Half wave.}$$

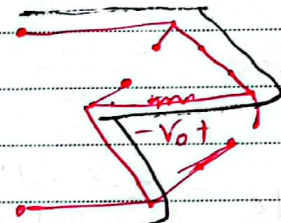
Peak Inverse Voltage.

For a Full Wave Bridge Rectifier,

$$+V_m - V_o = 0$$

$$\Rightarrow V_o = V_m$$

$$PIV \geq V_m$$

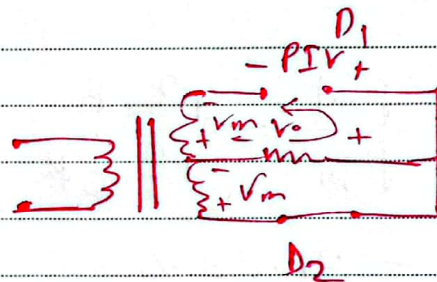


For Center Tapped:

$$+V_o - PIV + V_m = 0$$

$$\Rightarrow PIV = 2V_m$$

$$\Rightarrow PIV \geq 2V_m$$



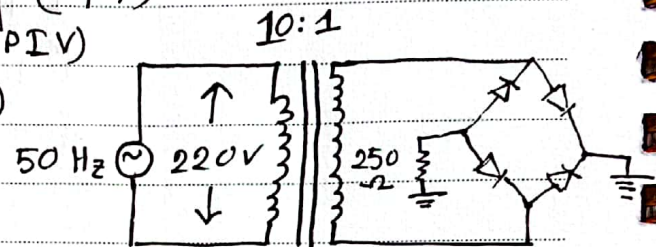
→ Better

Diode Rectifier Circuits (Numerical Problem)

For the circuit shown in figure, determine

- d.c. output voltage (V_{dc} or V_{av})
- Rectification efficiency ($\eta\%$)
- Peak inverse voltage (PIV)
- Output frequency (f_o)

Assume the diodes are real.



$$f_{in} = 50 \text{ Hz}$$

$$\frac{N_p}{N_s} = \frac{10}{1} = \frac{V_p}{V_s} = \frac{220V}{V_s} \Rightarrow V_s = 22V$$

$$\therefore V_s = V_i = 22V$$

$$(a) \quad V_{av} = \frac{2 V_m}{\pi}$$

$$V_{av} = V_{dc} = \frac{2 \times 31.11}{3.14} = 19.81 V \quad (\text{Ans})$$

We know,

$$V_{rms} = \frac{V_m}{\sqrt{2}} = 22V$$

$$\therefore V_m = 22\sqrt{2} V = 31.11 V$$

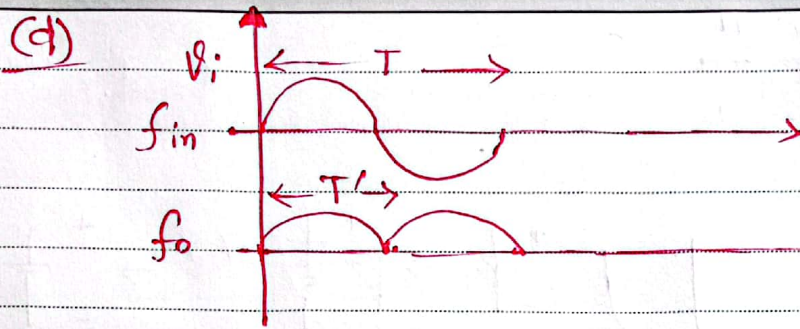
$$(b) \quad \eta = \frac{P_{out}}{P_{in}} = \frac{I_{dc}^2 R}{I_{rms}^2 R} = \frac{V_{dc}^2 / R}{V_{rms}^2 / R} = \frac{V_{dc}^2}{V_{rms}^2} = \frac{(19.81)^2}{(22)^2} = 31.08\%$$

(Ans.)

$$(c) \quad PIV \geq V_m$$

$$PIV \geq 31.11 \text{ volts}$$

Time period: Time taken by the waveform to repeat itself.



$$f = \frac{1}{T} \Rightarrow T = \frac{1}{f}$$

$$T' = \frac{1}{2} T$$

$$T = 2T'$$

$$\frac{1}{f_{in}} = 2 \times \frac{1}{f_o}$$

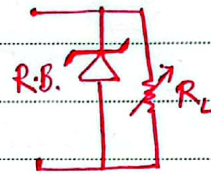
$$\Rightarrow f_o = 400 \text{ Hz}$$

(Ans)

Introduction to Zener Diode

* Zener diode is a special diode.

→ Voltage regulator সিএলসি used



এটাকে ভাঙা
Replace করবে

* Avalanche breakdown

lightly doped diode (Normal Diode)

অন্যদিকে Zener diode highly doped



* Zener breakdown

→ breakdown is achieved much earlier

↳ due to rupture of co-valent bonds

↳ tearing high E. field

See geeks4geeks for comparison

* High power dissipation capability

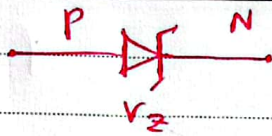
↳ (অবস্থা) breakdown
করেন সত্য কথা

* Properties will not degrade.

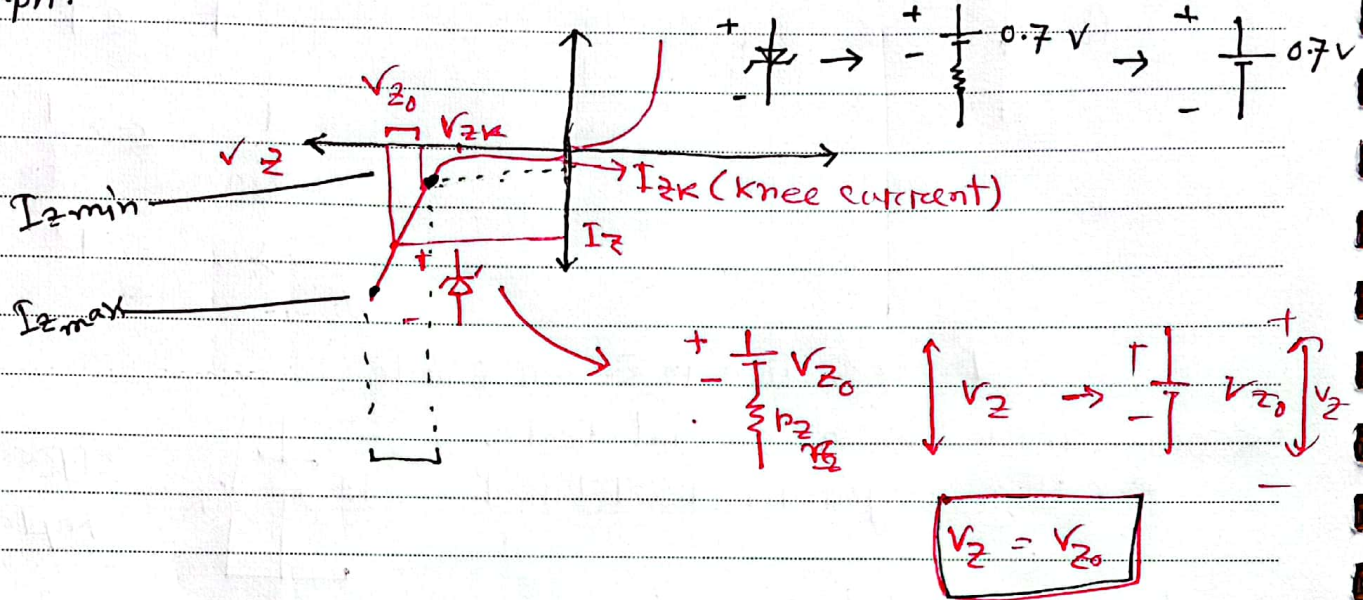
• www.corebrandtex.com

* Zener diode in f.b. সিএলসি ← নতুন ভাঙা সিএলসি সিএলসি

Symbol:



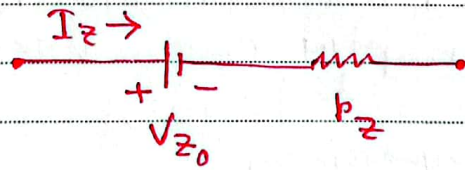
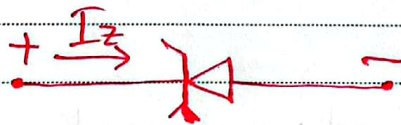
Graph:



- * For normal diodes: Breakdown voltage $> 6V$
- For Zener diodes: Breakdown voltage $< 6V$

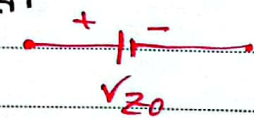
Depletion Layer: Zener Diode এর ডে প্লেশন (বাধা, নষ্টমান) আছে।

R.B.:



$$V_Z = V_{Z0} + I_Z R_Z$$

তবে আমরা R_Z avoid করি।



$$V_Z = V_{Z0}$$

Temperature Effects on Zener Voltage.

Temperature Co-efficient,

$$T_c = \frac{\Delta V_z / V_z}{T_1 - T_0} \times 100\% / ^\circ C$$

ΔV_z = change in zener voltage

V_z = Zener voltage at $25^\circ C$ (ROOM TEMP)

T_1 = new temp.

T_0 = Room, $25^\circ C$

Example: Calculate the change in zener voltage if zener voltage is 10V at $25^\circ C$ and new temperature is $100^\circ C$.
Temperature cof. (T_c) is $0.072\% / ^\circ C$.

$$\Delta V_z = ? \quad V_z = 10V \quad T_1 = 100^\circ C, \quad T_0 = 25^\circ C$$

$$T_c = 0.072\% / ^\circ C$$

$$\therefore 0.072 = \frac{\Delta V_z / 10V}{75} \times 100\% / ^\circ C$$

$$\Rightarrow \Delta V_z = 0.54 \text{ volt}$$

$$\therefore \text{New zener voltage: } V_{z_2} = 10 + 0.54 = 10.54V$$